

Daffa Narendra Hutapea

Data Scientist

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SUMMARY

Mathematics graduate from Universitas Indonesia and a Hactiv8 Full Time Data Science graduate, skilled in building and deploying machine learning and deep learning models (classification, clustering, computer vision, NLP, and retrieval-augmented LLM applications) using Python and SQL. Combines a strong analytical and statistical foundation with proven leadership and organizational experience from large-scale university events. Eager to apply data-driven problem solving and a growth mindset to create business impact as a Graduate Trainee.

EDUCATION

Hactiv8 Bootcamp

Full Time Data Science Program | Final Score: 96.4% | Honor Student

April 2026 – June 2026

Jakarta, Indonesia

Universitas Indonesia

Bachelor of Science (B.Sc.) in Mathematics | GPA: 3.45/4.00

July 2022 – February 2026

Depok, Indonesia

SMA Negeri 99 Jakarta

Natural Science Major

July 2019 – May 2022

Jakarta, Indonesia

WORK EXPERIENCE

PT Kimia Farma, Tbk

Big Data Analytics Intern

July 2024 – August 2024

Jakarta, Indonesia

- Gained hands-on experience in Data Warehouse process flow, Data Marts, Big Data concepts, and Data Pipeline management.
- Wrote SQL queries (including Common Table Expressions) and performed Create, Select, Update, and Delete operations on a Database Management System.
- Performed data analysis and visualization in Salesforce and built a sales dashboard in Google Looker Studio.
- Delivered a final project analyzing Kimia Farma's business performance for 2020–2023.

SKILLS & COMPETENCIES

General Skills: EDA, Machine Learning, Deep Learning, Computer Vision, NLP, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Unsupervised Learning, Statistical Analysis, Data Visualization

Programming Languages: Python, SQL, LaTeX

Libraries / Frameworks: Scikit-Learn, TensorFlow, Keras, XGBoost, imbalanced-learn, BERTopic, Transformers, LangChain, FastAPI, Gensim, Pandas, NumPy, Matplotlib, Seaborn

Modeling Algorithms: Regression, Decision Trees, Random Forest, KNN, SVM, Boosting (XGBoost), CNN, K-Means, PCA, Isolation Forest, LDA

Techniques: Feature Engineering, Hyperparameter Tuning, Cross-Validation, Transfer Learning, Image Augmentation, Topic Modeling, Text Embeddings, Semantic Search, Prompt Engineering, Dimensionality Reduction, Cluster Validation (Silhouette, Davies-Bouldin)

Tools / Deployment: Streamlit, Docker, Hugging Face Spaces, Vercel, ChromaDB (vector database), Groq API, Git, Google BigQuery, Power BI, Google Looker Studio, Elasticsearch, Kibana, Microsoft Office, Google Workspace

Soft Skills: Leadership, Teamwork, Problem Solving, Analytical Thinking, Communication

Languages: Indonesian (Native), English (Professional Working Proficiency; TOEFL ITP Equivalent: 570)

PROJECTS

FinChat: Retrieval-Augmented Q&A over SEC 10-K Filings (LLM / RAG) [Deploy]

July 2026

Built a Retrieval-Augmented Generation (RAG) chatbot answering plain-English questions over the SEC 10-K filings of 25 major U.S. companies, with every answer grounded in the source text and cited to prevent hallucination. Combined semantic search (BGE embeddings + ChromaDB) with hybrid XBRL retrieval and Python-computed financial ratios, generating answers via Llama 3.3 70B on Groq. Lifted FinanceBench accuracy from 20% to 50%, scored 93% on a capability gold set, and deployed it as an interactive web app.

Technology / Tools: Python, LangChain, ChromaDB, Hugging Face Embeddings (BGE), Groq (Llama 3.3 70B), edgartools (SEC EDGAR), XBRL, Next.js, FastAPI, Vercel, Hugging Face Spaces, Git

Land-Use Classification of Satellite Imagery (Computer Vision) [Deploy]

June 2026

Built a 5-class land-use classifier for Sentinel-2 satellite tiles (EuroSAT: AnnualCrop, Forest, Highway, Industrial, River) to enable large-scale land monitoring without manual photo-interpretation. Benchmarked a from-scratch CNN against a fine-tuned MobileNetV2 transfer-learning model trained on the same image-augmented data, reaching 97.1% test accuracy, and deployed it as an interactive web app.

Technology / Tools: Python, TensorFlow, Keras, MobileNetV2, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Docker, Hugging Face Spaces

Auto-Insurance Claim Fraud Detection [Deploy]

May 2026

Developed a binary classification model as a claims-screening tool that assigns a fraud risk score to each auto-insurance claim, helping investigators prioritize which cases to review. Benchmarked five algorithms (Decision Tree, Random Forest, KNN, SVM, XGBoost) within an imbalanced-learn pipeline using SMOTE and 5-fold stratified cross-validation, then tuned the best model (XGBoost) with GridSearchCV and a recall-leaning decision threshold, and deployed it as an interactive web app.

Technology / Tools: Python, Scikit-Learn, XGBoost, imbalanced-learn (SMOTE), Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Docker, Hugging Face Spaces

Topic Modeling of E-Commerce Reviews (NLP) [Deploy]

June 2026

Mined 5,400 Indonesian Tokopedia reviews (PRDECT-ID) to uncover what drives praise versus complaints. Benchmarked BERTopic (embeddings + UMAP + HDBSCAN) against LDA and K-Means baselines, modeled separately on positive and negative reviews. BERTopic led on coherence and diversity, surfacing 8 praise and 9 complaint themes with actionable business recommendations.

Technology / Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, BERTopic, Transformers, Gensim (LDA), Scikit-Learn, UMAP, HDBSCAN

ORGANIZATIONAL EXPERIENCE

Lecturer & Laboratory Assistant – Numerical Differential Equation

February 2025 – June 2025

Universitas Indonesia

Depok, Indonesia

- Taught and mentored 70+ second-year students per section in numerical programming using MATLAB and Octave.
- Designed new teaching materials, timelines, and graded problem sets, and evaluated student performance.

Laboratory Assistant – Numerical Methods

February 2024 – June 2024

Universitas Indonesia

Depok, Indonesia

- Instructed 50+ first-year students per section in Python and LaTeX and coordinated with 15+ fellow assistants.
- Created teaching materials and problem sets and evaluated student assignments.

Lecturer Assistant – Introduction to Data Science

February 2024 – June 2024

Universitas Indonesia

Depok, Indonesia

- Delivered supplementary lessons to 40+ students and evaluated their assignments to strengthen subject understanding.

HIPMI PT UI 2023/2024

September 2023 – August 2024

Staff of Creative Economy, MSMEs, and Startup Development

Depok, Indonesia

- Served as Head of Event Runner at Makarapreneur, a national business plan competition.
- Coordinated across the Visual Design, Public Relations, Operations, Logistics, and Competition divisions to deliver the Final Round and Gala Dinner.

Orientasi Kegiatan Kampus UI 2023 (OKK UI 2023)

April 2023 – August 2023

Vice Head of Sponsorship

Depok, Indonesia

- Approached 100+ corporations across Instagram, Email, and LinkedIn, securing sponsorship from 24+ companies (e.g. Gojek, Bank Indonesia, Frisian Flag, Telkomsel).
- Led a team of sponsorship staff and coordinated with the treasurer to fulfill sponsor deliverables and accountability reports.

ACHIEVEMENTS

- **Hacktiv8 Honor Student** – Ranked #1 in Batch HCK-040 with a final score of 96.4% across 3 phases; Honor Student status awarded for achieving a grade of 95% or higher.
- **Best Sponsorship Staff** – OKK UI 2023; promoted to Vice Head of Sponsorship for outstanding performance.
- **10th Highest Cumulative Score** – 2022 Natural Science cohort, SMA Negeri 99 Jakarta.

CERTIFICATIONS

Lembaga Bahasa Internasional (LBI), Universitas Indonesia

October 2022

English Proficiency Test (EPT) – Score: 570 (TOEFL ITP Equivalent), Grade: A

- Certificate No. 11959/UN2.F7.LBI/PDP.00.07.01/2022

HackerRank

July 2026

Problem Solving (Basic) – Skill Certification